VoxShield SIH 26052 PPT PDF Blueprint

Team Name: Team@ Required deck length: 6 slides total

Design direction

The deck must look like a finalist technical board, not a cinematic defence poster. Use clean finalist-style design:

- light grey or white background
- navy section headers
- white content cards
- thin grey borders
- green proof highlights
- amber next-step/risk highlights
- real workflow diagrams
- proof cards
- evidence board

Avoid:

- soldier face full-background
- tank/helicopter battlefield scenes
- huge glowing product title
- neon cyber dashboard look
- repeated fake waveforms
- unsupported dB claims

Slide 1: Title page

Purpose

Establish identity and seriousness.

Layout

Use official SIH header and footer. The slide body should have two zones:

- left: title and metadata
- right: clean system visual

Exact text

Title:

VoxShield

Subtitle:

AI speech enhancement for noisy defence communication

Metadata:

Problem Statement ID: 26052 Theme: Defence and National Security PS Category: Hardware Team Name: Team@

Proof badges:

ONNX export passed Pi 5 real-time benchmark Broad strict gate passed

Visual

Use a simple vector-style system visual:

headset mic -> VoxShield AI engine -> enhanced speech -> dashboard

No battlefield photo.

Slide 2: Idea title and proposed solution

Purpose

Explain the solution concept in five seconds.

Correct layout

Three-column board plus proof strip:

- left: problem context
- middle: proposed system workflow
- right: prototype outputs
- bottom: proof strip

Title

Idea title and proposed solution

Product heading

VoxShield: AI speech enhancement engine for noisy defence communication

Subtitle

Improves command intelligibility under vehicle, rotor, impulse and ambient noise using local real-time inference.

Left card: Problem context

Header:

Problem context

Rows:

Vehicle and engine noise - masks low-frequency speech cues

Rotor and wind noise - creates modulated nonstationary noise

Impulse noise - can hide short command words

Ambient field noise - reduces communication intelligibility

Bottom result:

Command words can get masked in high-noise communication.

Middle card: Proposed system workflow

Header:

Proposed system workflow

Supporting line:

Use a reference-noise path plus speech-preserving enhancement, then expose runtime health for the operator.

Workflow:

1. Primary mic - speech near mouth
2. Reference mic - environment noise

3. Streaming model - dual-channel deep filter
4. Retention guard - checks speech deletion risk
5. Enhanced output - audio plus telemetry

Lower boxes:

Validated software path: ONNX export, offline audits, failure analysis and controlled replay demo.

Hardware note: true physical validation needs synchronized raw dual-microphone capture.

Right card: Prototype outputs

Header:

Prototype outputs

Rows:

Enhanced speech - clearer command audio

Runtime telemetry - latency, bypass and health state

Evidence report - SNR, STOI, PESQ and deletion

Edge deployment path - ONNX model and Pi 5 benchmark

Bottom note:

Output is a prototype evidence path, not a claimed field-ready headset.

Bottom proof strip

Cards:

30.48 dB SNR - v15 broad strict

5.33 ms latency - Pi 5 mean for 16 ms frame

0.0 speech deletion - reported audits

Next: dual-mic validation - synchronized raw capture

Design notes

Do not use photo background. Do not make VoxShield huge. Do not use waveforms except as tiny optional line art. Do not claim final battlefield readiness.

Slide 3: Technical approach

Purpose

Show that the team built a full pipeline.

Layout

Use an architecture map with two layers.

Top layer: training pipeline

Data sources -> mixture builder -> training -> hard failure repair -> validation

Bottom layer: runtime pipeline

Primary mic + reference mic -> frame buffer -> ONNX streaming model -> enhanced audio -> dashboard/evidence report

Right proof rail:

1.1M parameters ONNX parity passed Pi 5 mean latency 5.33 ms 0.0 speech deletion in audits

Bottom amber note:

Laptop/Bluetooth demos are not proof of synchronized reference capture. Final field validation needs raw dual-mic hardware.

Text blocks

Data sources:

VCTK, DNS Challenge, MAD, ESC-50, AudioSet, RIRS, user recordings

Training:

SNR variation, reverb, impulse noise, nonstationary noise and hard-case replay

Runtime:

ONNX model, 16 kHz audio, 16 ms frame, dashboard telemetry

Slide 4: Feasibility and viability

Purpose

Show proof, risk and practical next step.

Layout

Top: four proof cards

Middle left: prototype readiness ladder

Middle right: hardware path diagram

Bottom: risk and mitigation table

Proof cards

Broad strict gate passed Pi 5 mean latency 5.33 ms ONNX export parity passed Speech deletion 0.0 in reported audits

Readiness ladder

1. Offline model validation - done
2. ONNX export - done
3. Pi 5 benchmark - done
4. Controlled replay demo - ready
5. Synchronized dual-mic field validation - next

Hardware diagram

Primary mic near mouth Reference mic facing noise USB audio interface Raspberry Pi 5 Headset/radio output Dashboard

Risk table

Risk	Why it matters	Mitigation
Laptop/Bluetooth not synchronized	weak live proof	raw USB dual-mic rig
Real noise differs from synthetic mixes	hard failures remain	field recordings and repair set
Suppression may damage speech	command content matters	STOI and speech deletion gates

Slide 5: Impact and benefits

Purpose

Make the problem human and mission-relevant.

Layout

Use before/after communication example.

Left:

Before VoxShield Noise masks important command words.

Example:

Hold position... left... unclear

Right:

After VoxShield Speech becomes clearer while command content is preserved.

Example:

Hold position. Left flank clear.

Bottom benefit cards:

Defence communication - improves command clarity in high noise

Emergency response - useful around sirens and crowd noise

Industrial safety - supports headset communication near machines

Edge operation - runs locally without cloud dependency

Metric card:

v15 broad audit: SNR 30.48 dB, STOI 0.938, PESQ 3.94

Slide 6: Research and references

Purpose

End with evidence and credibility.

Layout

Four-quadrant evidence board.

Quadrant 1: Data sources

VCTK, DNS Challenge, MAD, ESC-50, AudioSet targeted, RIRS NOISES, user recordings

Quadrant 2: Metrics

SNR, SI-SNR, STOI, PESQ, speech deletion, active level error, latency

Quadrant 3: Artifacts

v15 broad strict result v15 military audit ONNX export report Pi 5 verification failure report saved audio examples

Quadrant 4: Next hardware phase

dual USB mic rig raw synchronized capture field noise recordings headset/radio integration dashboard demo

Final line:

Current status: AI engine and edge runtime validated. Next phase: synchronized microphone hardware validation.

Final deck rules

Use fewer photos and more diagrams. Use fewer slogans and more proof. Keep every claim defensible. Make the project feel engineered, not imagined.